

T115 Beta Distribution Validation Documentation

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Introduction

This documentation will go over:

- Details of the beta distribution mapping process
- Details of the beta distribution validation process including the methods or processes used for validation, what information is included in the results, and an analysis of the outcomes.

Beta Distribution Mapping

- Rearrange the publicly available attrition rates over 9-year time windows (Sheet titled “17.1” in Excel file “2023 Section 17 – Completion Rates”) in a tabular format that is easy to process in R. Assume attrition rate is the sum of the percentage of students that re-enrolled but dropped out and the percentage of students that never came back after the first year. For example, for Attendance mode:

Category	Year	Completed	StillEnrolled	ReEnrolled	NeverCame	Attrition
Internal	2005	0.7664	0.0377	0.1267	0.0693	0.196
Internal	2006	0.7626	0.039	0.1318	0.0666	0.1984
Internal	2007	0.7645	0.0373	0.1316	0.0666	0.1982
Internal	2008	0.7687	0.0356	0.1351	0.0606	0.1957
Internal	2009	0.7667	0.0384	0.1391	0.0558	0.1949
Internal	2010	0.7577	0.0388	0.1427	0.0607	0.2034
Internal	2011	0.757	0.038	0.146	0.0591	0.2051
Internal	2012	0.7516	0.0416	0.1453	0.0615	0.2068
Internal	2013	0.7437	0.0428	0.1445	0.069	0.2135
Internal	2014	0.7409	0.0398	0.1501	0.0693	0.2194
Internal	2015	0.74	0.0378	0.1533	0.069	0.2223
External	2005	0.4666	0.0815	0.2489	0.203	0.4519
External	2006	0.4632	0.0736	0.2633	0.1998	0.4631
External	2007	0.4653	0.0797	0.2641	0.1908	0.4549
External	2008	0.4725	0.0757	0.266	0.1859	0.4519
External	2009	0.4838	0.0783	0.264	0.1739	0.4379
External	2010	0.4876	0.0767	0.2581	0.1775	0.4356
External	2011	0.4857	0.0745	0.2625	0.1774	0.4399
External	2012	0.4853	0.0757	0.2649	0.1741	0.439
External	2013	0.4688	0.0802	0.2611	0.19	0.4511
External	2014	0.4624	0.0716	0.2661	0.1999	0.466
External	2015	0.4639	0.0713	0.2707	0.1942	0.4649
Multi-modal	2005	0.7061	0.0431	0.1667	0.0841	0.2508
Multi-modal	2006	0.701	0.048	0.1736	0.0775	0.2511
Multi-modal	2007	0.7126	0.0487	0.165	0.0736	0.2386
Multi-modal	2008	0.7206	0.0468	0.1654	0.0672	0.2326
Multi-modal	2009	0.7164	0.0494	0.1735	0.0606	0.2341
Multi-modal	2010	0.726	0.0457	0.1701	0.0582	0.2283
Multi-modal	2011	0.7087	0.0464	0.1861	0.0588	0.2449
Multi-modal	2012	0.7156	0.0482	0.1787	0.0575	0.2362
Multi-modal	2013	0.7061	0.0522	0.177	0.0647	0.2417
Multi-modal	2014	0.7084	0.0476	0.178	0.066	0.244
Multi-modal	2015	0.7212	0.0403	0.171	0.0676	0.2386

- Take attrition column for each category. For example, for “Internal” attendance mode:

Category	Year	Attrition
Internal	2005	0.196
Internal	2006	0.1984
Internal	2007	0.1982
Internal	2008	0.1957
Internal	2009	0.1949
Internal	2010	0.2034
Internal	2011	0.2051
Internal	2012	0.2068
Internal	2013	0.2135
Internal	2014	0.2194
Internal	2015	0.2223

- Calculate alpha (α) and beta (β) parameters using Method of Moments estimators:

Let m = mean attrition rate and v = variance in attrition rate (calculated using R built-in mean() and var() functions).

$$\alpha = m \times \left(\frac{m \times (1 - m)}{v} - 1 \right)$$

$$\beta = (1 - m) \left(\frac{m \times (1 - m)}{v} - 1 \right)$$

- Resulting parameters that are fed into dbldatagen:

Attendance_Mode	Alpha	Beta
Internal	353.4117	1371.542
External	888.0678	1082.947
Multi-modal	830.1189	2627.531

Beta Distribution Validation – Public Attrition Rates

Please see **Public Data -> Attendance Mode.xlsx**, **Broad FOE.xlsx**, **Course Admission Load Category.xlsx**, and **Course Level.xlsx** for the results.

KS and AD tests

The purpose of performing KS and AD tests is to use formal statistical tests to investigate the hypothesis of whether the beta distribution adequately fits the public attrition rates.

- Take attrition rates, alpha and beta parameters for each category. For example, for “Internal” attendance mode:

Year	Attrition	Alpha	Beta
2005	0.196	353.4117	1371.542
2006	0.1984		
2007	0.1982		
2008	0.1957		
2009	0.1949		
2010	0.2034		
2011	0.2051		
2012	0.2068		
2013	0.2135		
2014	0.2194		
2015	0.2223		

- Run KS and AD tests using these values as inputs into R’s built-in functions:

```
ks.test(Attrition, "pbeta", shape1 = Alpha, shape2 = Beta)
```

```
ad.test(Attrition, null = function(z) pbeta(z, Alpha, Beta))
```

- Results:

Category	KS_pvalue	KS_D	KS_alt	AD_pvalue	AD_An
Internal	0.70060133	0.20006865	two-sided	0.74848668	0.49467501
External	0.80753068	0.1929258	two-sided	0.79990165	0.44492812
Multi-modal	0.99599954	0.12368481	two-sided	0.99703286	0.16932048

- All results show p-values that are significantly larger than 0.05 for both KS and AD tests. This suggests that there is no evidence against the idea that a beta distribution is a good fit for the publicly available attrition rate data.
- Sample sizes that contributed to attrition rates in publicly available data is unknown. Input sample size for KS and AD tests performed on publicly available data are always the number of commencing years (11). This is an incredibly small sample size. Therefore, the results of

the KS and AD tests are likely unreliable and require visualisations and parameter checks to reasonably determine goodness-of-fit.

Parameter Stability

The purpose of testing for parameter stability is to check whether the estimated parameters are meaningful and independent of noise or defects in the samples such as sample sizes – which was a factor that limited the reliability of the KS and AD test results.

- Calculate alpha and beta parameters using Maximum Likelihood Estimation (MLE), Minimum Detectable Effect (MDE) across the Interquartile Range (0.25, 0.5, 0.75), and MDE using only the median. These were calculated in R and compared with the original alpha and beta parameters calculating using the Method of Moments (MoM) to compare parameter values. If alpha and beta remain somewhat similar across all methods, the parameters are stable:

Attendance _Mode	Alpha_ MoM	Beta_ MoM	Alpha_ MLE	Beta_ MLE	Alpha_ MDE_IQR	Beta_ MDE_IQR	Alpha_ MDE_ Median	Beta_ MDE_ Median
Internal	353.41	1371.5	394.504	1531	351.4961	1364.108	27.258	105.783
External	888.07	1082.9	977.334	1191.8	531.1564	647.7146	4505.6	5494.36
Multi- modal	830.12	2627.5	914.309	2894	914.915	2895.932	28.148	89.0958

- Overall, all methods had similar values for alpha and beta except for MDE using the median. However, MDE using the Interquartile Range had similar results to MLE and MoM. This is likely due to the incredibly small sample size making the median an unreliable measure of centre.

Histogram

- A histogram of attrition rates and a line plot of the mapped beta distribution was developed for each category to visualise the goodness-of-fit of the beta distribution to the attrition rates.
- For histograms that have a single peak, the beta distribution maps the attrition rate quite well.
- For histograms that have more than one peak, the beta distribution estimates the best shape with a single peak.
- Overall, it appears the beta distribution fits to the attrition rates as best as it can.

CDF

- The empirical CDF (black dots) and the beta distribution CDF (red line) was plotted for each category to visually confirm the goodness-of-fit of the beta distribution in regard to cumulative probability.
- All plots show that the empirical and beta CDF are somewhat similar which suggests that the beta distributions are a decent match to the data's cumulative behaviour.

QQ

- QQ plots for each category were plotted to visually confirm the goodness-of-fit of the beta distribution in regard to quantiles.
- All plots have points that follow somewhat closely to the line which suggests that the beta distribution fits the data quite well across a range of quantiles.

Beta Distribution Validation – Synthetic Data

Please see **Synthetic Data -> Attendance Mode.xlsx, Broad FOE.xlsx, Course Admission Load Category.xlsx**, and **Course Level.xlsx** for the results.

KS and AD tests

- Most results have p-values that are less than 0.05 which means the attrition rates in the synthetic data do not follow the beta distributions. Since the beta distributions had a satisfactory fit with the publicly available data, this means that the attrition rates in the synthetic data do not follow the same statistical trends as the public attrition rates.
- Input sample size for KS and AD tests performed on the synthetic data are always the number of years included in the data (6). This is an incredibly small sample size. Therefore, the results of the KS and AD tests are likely unreliable.

Histogram

- A histogram of attrition rates and a line plot of the mapped beta distribution was developed for each category.
- Many categories had an even distribution of attrition rates – which none of the publicly available attrition rates followed and therefore, is likely not a realistic depiction of attrition rates.

CDF

- The empirical CDF (black dots) and the beta distribution CDF (red line) was plotted for each category.
- Most plots show a severe divergence between the empirical and beta CDF – which means that the attrition rates in the synthetic data do not match the beta distribution.

QQ

- QQ plots for each category were plotted.
- Most plots show a severe divergence between the points and the line – which mean that there is a significant mismatch between the sample quantiles and the theoretical beta quantiles.

Limitations and Considerations

- Not all categories in the publicly available data could be mapped one-to-one with a category in the synthetic data. For example, the course levels in the publicly available data only span Bachelor's levels while the synthetic data spans up to Postgraduate levels as well. Therefore, not all beta distributions that were mapped were validated with the synthetic data.
- For Attendance Mode, it was assumed that the categories "Internal", "External", and "Multi-modal" were mapped one-to-one to "On-campus", "Online", and "Mixed / Other" respectively. This may not be correct.
- Since the beta distribution parameters were calculated using the public data's mean and variance, the observed mean and variance (from the public data) and the fitted mean and variance (from the beta distribution) are the same. Therefore, the observed and fitted mean and variance were omitted from the results in the public data validation.